MFE FORMWORK MR11 SYSTEM

Software Requirements Specification

Version 1.0

Prepared for

MFE Formwork

Prepared by

MyIntellX project team

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Document control

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1. Summary

Version 1.0 meets 47 functional requirements across six modules, plus the non-functional requirements in section 5. Each requirement was checked against the source code at commit `0cc8c36`.

2. Purpose and scope

The system replaces the manual, workbook-by-workbook MR11 Master Schedule. Seven departments upload their usual Excel workbooks; the system merges them into one project-by-project schedule, a CEO dashboard and an administration console.

In scope (version 1.0)	Out of scope
Sign-in, password change, accounts managed by administrators	Self-registration, single sign-on
Workbook upload and viewing for BD, Finance, Shell Plan & Design, Planning, Production, Dispatch (Local and Overseas)	Editing department data in the browser
MR11 Master Schedule with filters, export, and admin row delete and restore	Email or push notifications
Series, approval and monthly dispatch history	Mobile app
CEO dashboard	ERP or accounting integration
Admin console	

3. Stakeholders and user roles

Role	Uses the system to
CEO / executives	Read the CEO dashboard and the MR11
Business Development (BD)	Upload the master project list (contract, area, price, delivery)
Finance	Upload payment status and amounts received
Shell Plan and Design	Upload consultant drawing status, approvals and design completion
Planning	Upload processing series and cumulative processed quantity
Production	Upload fabrication output
Dispatch	Upload Local and Overseas shipment and delivery data
System administrator	Manage users and roles; delete or restore MR11 rows

4. Permissions

Capability	ADMIN	CEO	Department roles
Sign in, change own password	Yes	Yes	Yes
View MR11, export to Excel, sync MR11	Yes	Yes	Yes
Delete or restore MR11 rows	Yes	No	No

Capability	ADMIN	CEO	Department roles
View, upload and download a department workbook	All departments	No	Own department only (SHELLPLAN and DESIGN share one workbook)
CEO dashboard	Yes	Yes	No
Admin console	Yes	No	No

5. Functional requirements

5.1 Authentication and accounts (AUTH)

ID	Requirement
AUTH-01	A user signs in with email and password. The email is matched without regard to case or surrounding spaces.
AUTH-02	A wrong email shows "No account found for this email"; a wrong password shows "Wrong password".
AUTH-03	A successful sign-in issues a session token valid for 7 days.
AUTH-04	A user still on the default password is prompted to change it.
AUTH-05	A user can change their password. The new password must be at least 8 characters, differ from the current one, and not be the default password.
AUTH-06	An expired or invalid session signs the user out and returns them to the sign-in page.
AUTH-07	Each page opens only for users allowed to see it; others are sent to the MR11 page. The server enforces the same rules on its data.

5.2 Department workbooks (DEPT)

ID	Requirement
DEPT-01	Each department page shows its active workbook as a spreadsheet, keeping values, fill colours, font colours and merged cells.
DEPT-02	An authorised user uploads an .xlsx file of up to 25 MB; it becomes the department's active version.
DEPT-03	Shell Plan and Design share one combined workbook ("Shell Plan & Design").
DEPT-04	Dispatch keeps two files, Local and Overseas. An upload replaces only its own file. Without a stated part, the system decides by file name, then by columns (ETD / ATD columns mean Overseas).
DEPT-05	Every upload is saved to the database before it becomes active. If saving fails, the uploader sees an error and the previous version stays active.
DEPT-06	Every upload is logged with user, department and file name, and triggers a rebuild of the MR11.
DEPT-07	Day headers stored as Excel date numbers display as dates.
DEPT-08	Planning and Production pages show series history; the Shell Plan & Design page shows approval history; the Dispatch page shows m² dispatched per month.

5.3 MR11 Master Schedule (MR11)

ID	Requirement
MR11-01	BD is the master list: every MR11 row starts from a BD row with a value in at least one cell.
MR11-02	Rows from other departments are matched to a BD row by project (short name first, then project number), stream, and the closest font and fill colour.
MR11-03	A project number may carry the stream as its last digit (251242 = project 25124, stream 2).
MR11-04	Finance fills Advance Received / Payment Status, Actual Received and Payment Date.
MR11-05	Shell Plan fills its status and approved date; Design fills Formwork Design Status, completion date per level and Total Quantity Ordered m².
MR11-06	Design completion dates are coloured: dark green = completed and passed, yellow = completed but still ahead, red = not completed.
MR11-07	Total Processed is the sum of Planning's Cumulative Processed (Project) for the row's colour blocks. Processed Date is the upload date on which that total last changed.
MR11-08	Total Produced is the sum of Production's Cumulative Produced (Project). Produced Date is the latest daily column with output.
MR11-09	Total Dispatch, Dispatch Date, Formwork Quantity Sailed (m²) and ETD/ATD come from the Dispatch files. Local projects show "Delivered:" dates instead of sailing dates.
MR11-10	A project is Local when it is in Peninsular Malaysia; Sabah, Sarawak, Labuan and other countries are Overseas, unless a Dispatch file already lists it.
MR11-11	Blue-font BD rows (Re-Design Only) stop at Planning; their Production and Dispatch columns show "-".
MR11-12	Monthly columns Jan-26 to Dec-27 show actual m² dispatched for ended months and the forecast for the current and later months, with yearly totals.
MR11-13	Rows of the same project and stream with the same fill colour share merged Shell Plan and Design cells.
MR11-14	Rows sort by project, stream, unfilled before filled, fill colour, then font colour (black first).
MR11-15	Users can search, filter rows, show or hide column groups, zoom and use full screen; the page remembers these settings per browser.
MR11-16	A department user's own columns are highlighted in the department colour.
MR11-17	Export produces an .xlsx file with the two-row header, row colours, number formats and merged cells.
MR11-18	An administrator can delete a row from the MR11 and restore it later; a deleted row stays out of every rebuild until restored.
MR11-19	Any signed-in user can sync, which rebuilds the MR11 from every department's latest workbook.

5.4 Pricing and market data (PRICE)

ID	Requirement
PRICE-01	Selling Price (USD) combines BD's USD price, MYR price and any other-currency price, converted at today's rate.
PRICE-02	LME Adjusted (USD) = (today's LME aluminium cash price − the row's LME Rate) ÷ 1000 × 21.5 for "Variable" rows; 0 for "Fixed"; "Check" for "Freeze".
PRICE-03	Final Selling Price (USD) = Selling Price + Props, WPB, Waler, Acc + Aluminium Weight Adjusted + Freight Adjusted + LME Adjusted.

ID	Requirement
PRICE-04	Prices are recalculated each time the MR11 is read or exported, not only on upload.
PRICE-05	Exchange rates come from Bank Negara Malaysia, with open.er-api.com and the European Central Bank (Frankfurter) as fallbacks; LME prices come from Westmetall. Both are cached for one hour, and the source is shown on screen.
PRICE-06	Other-currency names are recognised by ISO code, English name or symbol (for example SGD, "Singapore Dollar", S$).

5.5 CEO dashboard (CEO)

ID	Requirement
CEO-01	The dashboard shows stage totals and per-row figures from the latest MR11, filtered by region and month.
CEO-02	A month pipeline shows, per month, the BD forecast, Design, Processed, Produced and Dispatched m². Blue-font rows are labelled Re-Design Only.
CEO-03	A pricing panel shows average selling price per m² with today's LME price and exchange rate.

5.6 Administration (ADMIN)

ID	Requirement
ADMIN-01	An administrator lists all users with their roles.
ADMIN-02	An administrator creates a user with email, name, password and a starting profile.
ADMIN-03	An administrator changes a user's roles, by profile or role by role.
ADMIN-04	An administrator deletes a user. The root account admin@mfeformwork.com cannot be deleted or lose its ADMIN role.

6. Non-functional requirements

Area	Requirement
Availability	Runs as serverless functions on Vercel (Singapore) or a container on Azure App Service. Health check at `/health`; database check at `/api/health/db`.
Data durability	Supabase PostgreSQL is the shared source of truth. Without a database the system runs from memory and loses data on restart (demo mode only).
Consistency	Each server instance reloads users, workbooks and the latest MR11 from the database before serving.
Security	Passwords stored hashed; every API except sign-in and health checks requires a signed token; role checks run on the server.
Performance	A BD upload with rebuild completed in 0.35 s in the pre-UAT test (sample data, local server).
Usability	Comma thousands separators; 3 decimal places for decimal numbers; payment amounts in dollars.
Time zone	"Today" and month boundaries follow Malaysia time (Asia/Kuala_Lumpur).

Area	Requirement
Browser support	Current desktop Chrome, Edge, Firefox and Safari.
File limits	Upload up to 25 MB per workbook; request body up to 100 MB.

7. Assumptions and constraints

- Departments keep their workbook layouts and header names. A renamed header may stop a column from mapping; the mapping list is in `server/config/mr11.config.ts`.
- Colour conventions (font and fill) carry meaning and stay consistent between departments; small shade differences are tolerated.
- Month columns cover January 2026 to December 2027; later years need a configuration change.
- Market data depends on public services (Bank Negara Malaysia, Westmetall) staying available.

8. Glossary

Term	Meaning
MR11	The master schedule that tracks every project from contract to dispatch
Stream	A sub-package of a project, taken from the project number or Stream column
Series	A batch of formwork processed (Planning) or fabricated (Production)
Font / fill colour	Row text and background colours; they mark blocks, plants and Re-Design Only rows
LME	London Metal Exchange aluminium price
ETD / ATD	Estimated / actual time of departure
RBAC	Role-based access control